



ICMR Center for Advanced Research 1st Annual Progress Review Meeting 2026

**Project Title: Implementing a Multifaceted, Comprehensive,
Appropriate Care Package for People with Advanced and Terminal
Illnesses Needing ICU Support (MATHRU)**

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Project ID: CAR-2024-01-000500/F2/F3

**Reporting Period: 15 September 2025 – 31 August 2026 (Sites – KMC,
SJMC)**

Section 1: Project Overview

To build a scalable and replicable model for palliative care **delivery** in the ICUs contextualised to India underpinned by implementation science research

Co-design of the Intervention Package
[Multifaceted Comprehensive Appropriate Care Package – MCACDP]

Implementing in two ICUs (O/C)

where PC is routinely delivered

Phase 1

Phase 2

Implementing in four ICUs where PC access is present but PC not routinely delivered in ICUs

Phase 3

Implementing in four ICUs where no PC access is present

Phase 4

Developing a Technical Resource Hub for PC delivery in Indian ICUs

Phase 5

6 Realist Reviews
Each review findings harmonised through multiprofessional stakeholder interviews
Delphi + NGT
MCACP Manual

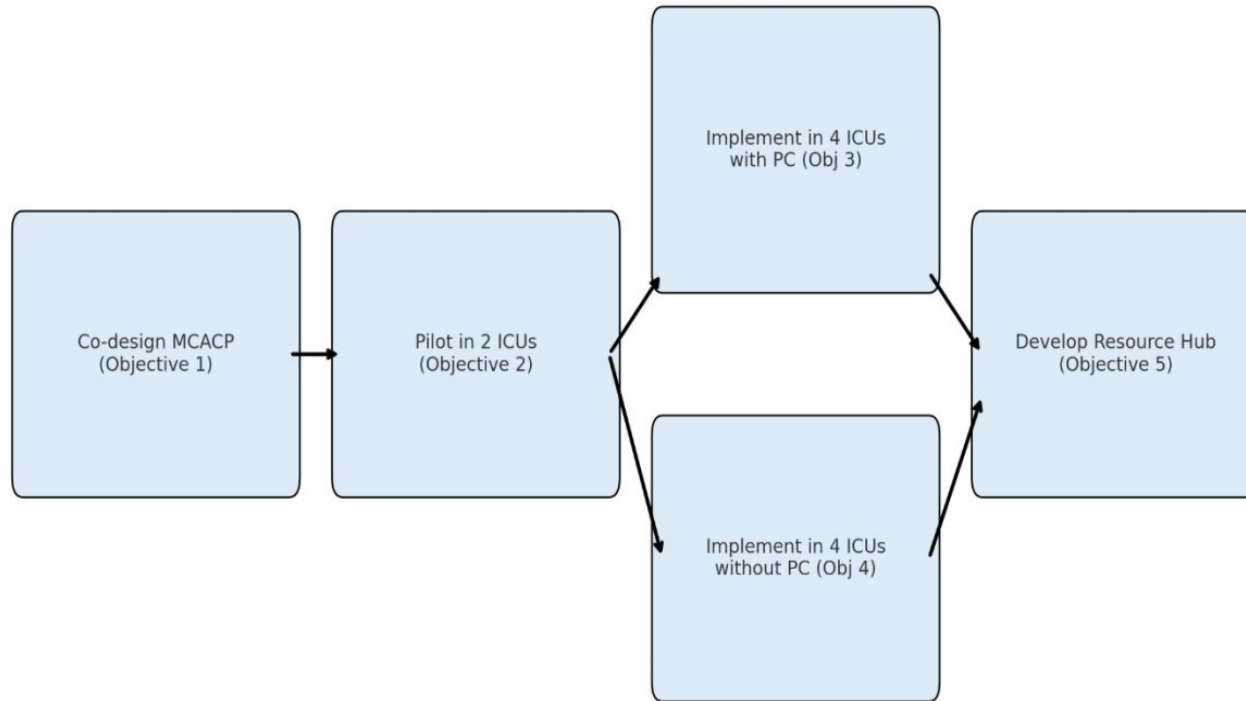
Mixed-methods study (observational + Qualitative)
Multiprofessional Patients/Families
MCACP refinement
Impl/Clin outcomes

Mixed-methods study (observational + Qual + Quasi Experimental)
Multiprofessional Patients/Families
MCACP refinement

Mixed-methods study (observational + Qual + Quasi Experimental)
Multiprofessional Patients/Families

Centralized, public-facing resource and learning hub containing protocols, templates, training modules, policies, legal framework, website etc.

Section 1: Project Overview



10 Clinical Sites
2 with PC delivery in ICU
4 with PC access but no
delivery
4 with no PC access

7 Technical Sites
Health economics
Bioethics
Public Policy
Health Law

- Co-design and evaluate the Multifaceted Comprehensive Appropriate Care Package (MCACP)
- Integrate palliative care, communication and shared decision-making into ICU workflows
- Improve quality of dying, family satisfaction and care planning
- Reduce avoidable treatment burden, expenditure and resource use
- Test implementation across contrasting Indian ICU contexts
- Develop a nationally scalable, modular knowledge translation and implementation support platform

Section 2: Research Question and Objectives

Research Question

How can a Multifaceted, Comprehensive, Appropriate Care Package (MCACP) for people with advanced and terminal illnesses needing ICU support be co-developed, implemented, and iteratively refined across diverse Indian ICU contexts?

What is its impact on contextual adaptability, acceptability, implementation feasibility, and patient, family, and system-level outcomes such as quality of dying, family satisfaction, resource utilisation, and cost?

Objectives	Work packages
1. Co-develop the MCACP intervention through iterative, participatory design	6 realist reviews – findings harmonised through stakeholder interviews, Iterative prototyping – Delphi + NGT, reporting TIDieR checklist, Manual dev
2. Pilot the MCACP intervention in 2 ICUs with palliative care delivery	Prospective observational - 100 ICU episodes/site (2 sites) Qualitative interviews - Multi-professional* + family caregivers + public representatives
3. Expand implementation to four ICUs with existing palliative care access but no delivery	Mixed methods – implementation planning + Mixed methods – implementation evaluation + Quasi-experimental – implementation effectiveness + Qual (ecosystem)
4. Extend implementation to four ICUs without palliative care access	Mixed methods – implementation planning + Mixed methods – implementation evaluation + Quasi-experimental – implementation effectiveness + Qual (ecosystem)
5. Develop a Technical Resource Hub for supporting PC delivery in ICUs Across India	Co-developing manuals, templates, courses, framework, website, videos, infographics, SOPs, case studies, patient decision aids etc – nationally accessible

In objective 2,3 and 4 – refine implementation and intervention and evaluate effectiveness after each objective – iteratively informs next steps

* Multiprofessional – ICU Physicians, Nurses, and SW, PC Physicians and Hosp Admin

Section 3: Methodology in brief

Study Design	Study Area	Sample Size	Implementation Plan
Objective 1: Co-develop the MCACP intervention through iterative, participatory design - MRC complex-intervention framework, participatory mixed-methods co-design			
1.1. Realist Review (6) – total 466 studies	Stakeholders from 10 participating clinical sites, representing open, closed and semi-open ICUs with varying palliative-care access	FGD Groups: 5 (intensivists, palliative-care physicians, nurses, allied-health professionals, administrators, technical experts and family caregivers) Participant per FGD Group: 5-10	Independent Syst Literature Search FGD per review, conduct stakeholder consultations (ongoing on site) and NGT Evidence contextualisation Initial Program Theory define seven MCACP components
1.2. Consensus (Two-step Delphi + NGT)		Delphi: 50 participants NGT: 10-15 participants	Conduct 2-Step Delphi process using the Conducting and REporting of DELphi Studies (CREDES) approach
1.3. Prototyping and Usability testing		8-10 ICU providers	
1.4. Reporting			prepare manual, SOPs, algorithms, templates, educational resources and workflows → undertake usability testing → freeze the version-controlled initial MCACP package
Objective 2: Pilot the MCACP intervention in 2 ICUs with palliative care delivery			
2.1. Prospective Observational	KMC Manipal: open consultant-based ICU model. SJMCH Bengaluru: closed intensivist-led ICU model with established palliative-care access	Site 1: 100 ICU Episodes Site 2: 100 ICU Episodes	Screen eligible ICU patients → collect baseline clinical, communication, family-experience, cost and resource-use data → orient and train ICU teams → implement the seven MCACP components → monitor adoption, fidelity, AIM, IAM and FIM → conduct interviews and workflow observations → document adaptations → implement adaptations and evaluate fidelity and effectiveness
2.2. Qualitative		Site 1: 36 Site 2: 30	

Section 3: Methodology in brief

Study Design	Study Area	Sample Size	Implementation Plan
Objective 3: Expand implementation to four ICUs with existing palliative care access but no delivery			
3.1. Qualitative	ICU Ecosystem pre-implementation readiness assessment across AIIMS Delhi, CMC Vellore, Tata Memorial Hospital Mumbai and Bharati Vidyapeeth Hospital Pune	25 multidisciplinary stakeholders	Realist-informed interviews guided by CFIR and NPT; assess beliefs, readiness, organisational culture, leadership, legal concerns, resources and barriers/facilitators
3.2. Mixed – Methods	Implementation planning	80 observed participants: 20/ICU; 25 qualitative participants	Map workflows and stakeholders; assess infrastructure and implementation climate; identify champions; tailor implementation strategies; train clinical and research teams
3.3. Mixed – Methods	Implementation evaluation	80 observed participants: 20/ICU; 25 qualitative participants	Implement MCACP; assess reach, adoption, fidelity, acceptability, feasibility and normalisation; conduct learning cycles; document adaptations using TIDieR
3.4. Quasi-Experimental	Implementation effectiveness	150 patients/ICU = 600 pre-intervention and 600 post-intervention	Compare pre-post quality of dying, family satisfaction, symptom control, care cost, ICU resource utilisation, palliative-care referral and preference-concordant care
Objective 4: Extend implementation to four ICUs without palliative care access			
4.1. Qualitative	ICU Ecosystem pre-implementation assessment across VMMC-Safdarjung Delhi, Aster CMI Bengaluru, Yashoda Hospital Ghaziabad and P. D. Hinduja Hospital Mumbai	25 multidisciplinary stakeholders	Assess organisational readiness, unmet palliative-care needs, workforce capacity, referral barriers, ethical/legal concerns and local implementation context
4.2. Mixed – Methods	Implementation planning	80 observed participants: 20/ICU; 25 qualitative participants	Establish basic palliative-care capacity; identify and train champions; develop referral pathways; adapt MCACP tools and workflows for ICUs without specialist services
4.3. Mixed – Methods	Implementation evaluation	80 observed participants: 20/ICU; 25 qualitative participants	Implement context-adapted MCACP; provide mentoring and implementation support; evaluate reach, fidelity,

Section 3: Methodology in brief

Study Design	Study Area	Sample Size	Implementation Plan
Objective 5: Develop a Technical Resource Hub for Supporting PC delivery in ICUs Across India			
Iterative co-design and knowledge-translation platform development	Nationwide government and private ICU networks, including district hospitals, teaching hospitals and palliative-care-integrated centres	Determined by content and setting diversity; test in at least one district hospital and one tertiary centre; blended-learning pilot in 2–3 sites	Organise and peer-review MCACP resources → design an open-access multilingual platform → upload manuals, SOPs, templates, videos and training modules → pilot online/offline delivery → align with ICMR/DGHS/NABH/NHM/NHA → establish maintenance, handover and uptake/outcome metrics

Section 4: Interim Modifications

NO CHANGE TO THE SANCTIONED OBJECTIVES, STUDY DESIGN OR 60-MONTH DURATION

GANTT CLARIFICATION

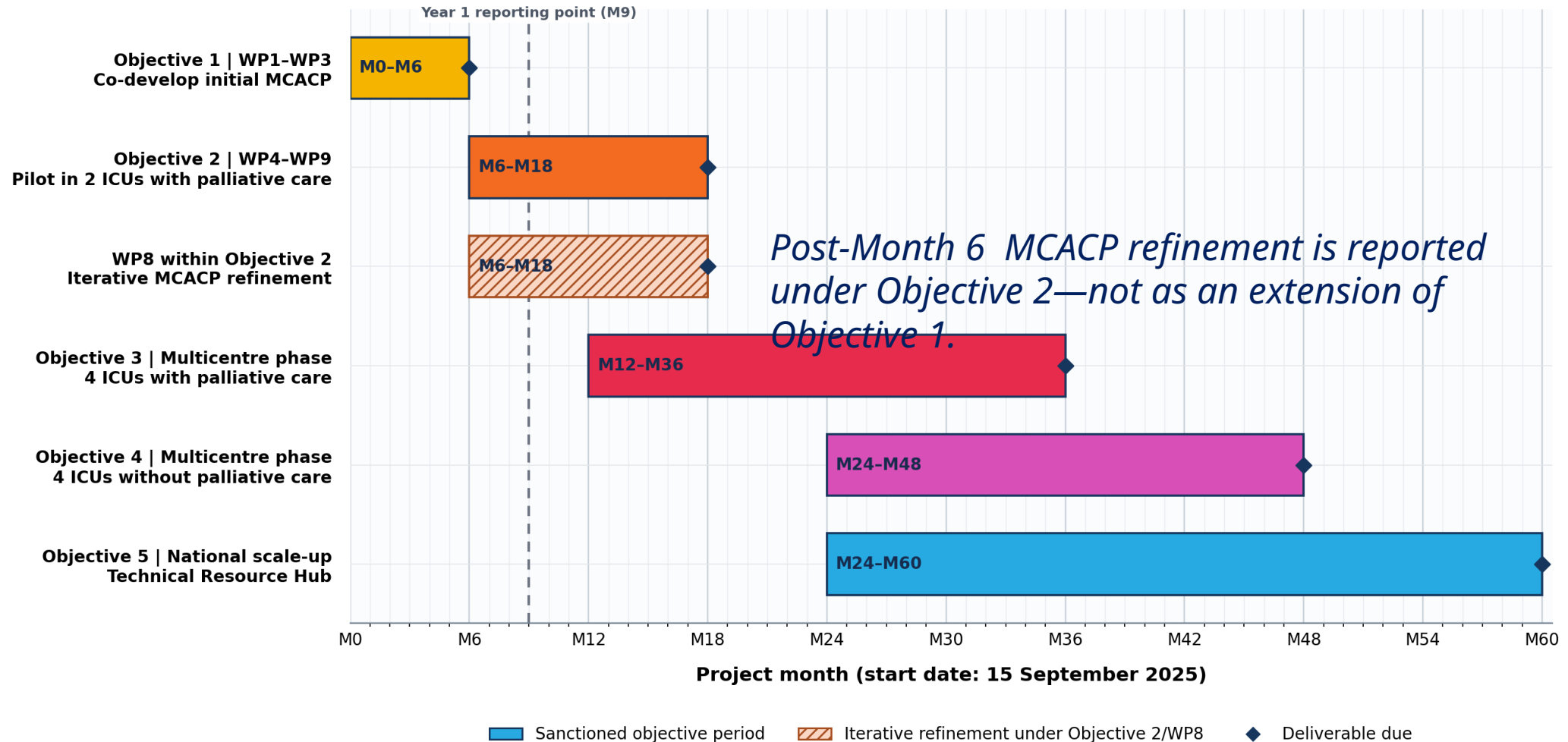
- The approved schedule retains Objective 1 at M0–M6 and Objective 2 at M6–M18.
- The Year 1 report displayed Objective 1 through M18 because co-development-related refinement continues during piloting.
- This continuing activity is now shown correctly under Objective 2/WP8: iterative MCACP refinement.

OPERATIONAL ADAPTATIONS WITHIN THE APPROVED METHODOLOGY

- MCACP manual treated as a living document refined through frontline feedback and PDSA cycles.
- Site-specific workflow adaptation for open and closed ICU models.
- Multicentre readiness work initiated in parallel to avoid delay in Objective 3.

Section 5: Timeline/Gantt Chart

Corrected Gantt Chart: ICMR CAR Grant (60-month sanctioned timeline)



Clarification: Objective 1 remains M0-M6. Co-development-related refinement after M6 is shown under Objective 2/WP8; no sanctioned end date is changed.

Section 6: Key Results – Objective 1

Six Linked Realist Syntheses and Stakeholder Consultation

Shared programme-level methods		Result
Unique papers screened across the programme		10,484
Stakeholder consultation		5 FGDs
Stakeholder participation		57 unique participants
Overall status		Completed
Intervention component synthesised	Evidence included in component synthesis, n	Review status
Prognostication	78	Completed
Communication	71	Completed
Shared decision-making	86	Completed
Physician consensus and multidisciplinary working	130	Completed
Care planning	102	Completed
Care delivery	79	Completed

Section 6: Key Results – Objective 1

Prognostication: Key findings (domains and themes) contributing to MCACP after evidence synthesis and evidence contextualisation after stakeholder interviews		
Domain/theme	Evidence synthesis and stakeholder contextualisation	Contribution to MCACP
Early recognition	Prognostic uncertainty commonly delays palliative-care referral and goals-of-care discussions. Stakeholders supported need-based identification rather than diagnosis-only triggers.	Simple screening triggers at admission and after deterioration, treatment failure, prolonged ICU stay or a major decision point
Dynamic prognostication	Prognosis changes over time and cannot be determined reliably by a single score or assessment.	Structured serial assessment incorporating diagnosis, illness trajectory, reversibility, organ support, frailty and treatment response
Clinical judgement and prediction tools	Prognostic scores may improve consistency but can create false certainty and should not define “futility.”	Scores used only as decision-support prompts; never as automatic criteria for treatment limitation
Multidisciplinary interpretation	Differences between intensivists, primary physicians and other specialists can produce conflicting messages and delayed decisions. Nurses often hold important bedside information but remain peripheral to formal deliberation.	Multidisciplinary prognostic review, explicit nursing input, physician-consensus process and identification of one communication lead
Managing uncertainty	Families and clinicians need honest information without premature certainty. Repeated discussions are often necessary as the illness evolves.	Document best-case, most-likely and worst-case scenarios, key uncertainties, time-limited treatment trials and predefined reassessment points

Section 6: Key Results – Objective 1

Communication: Key findings (domains and themes) contributing to MCACP after evidence synthesis and evidence contextualisation after stakeholder interviews

Domain/theme	Evidence synthesis and stakeholder contextualisation	Contribution to MCACP
Timely initiation	Communication is often delayed until deterioration or crisis. Families may receive important information only after treatment escalation.	Early communication triggers linked to prognostic concern, treatment failure, ICU admission and major decision points
Team consistency	Different clinicians may provide fragmented or conflicting explanations. Stakeholders emphasised the need for team alignment before family discussions.	Pre-meeting multidisciplinary huddle, agreed clinical recommendation and identification of one communication lead
Explaining prognosis and uncertainty	Technical language, excessive certainty and inconsistent descriptions of prognosis reduce understanding and trust.	Plain-language explanation using best-case, most-likely and worst-case scenarios, including what remains uncertain
Family-centred communication	Families require acknowledgement of emotions, opportunities to ask questions and repeated discussions as their understanding evolves.	Structured family-meeting guide incorporating empathy, active listening, emotional support, repetition and follow-up
Values and preferences	Communication frequently focuses on procedures rather than the patient's values, acceptable outcomes and prior wishes.	Explicit exploration of what matters to the patient, acceptable trade-offs, advance directives and family knowledge of prior preferences
Checking understanding	Information provision does not ensure comprehension, particularly during emotional distress or when language and health-literacy barriers exist.	Teach-back, interpreter support, culturally and linguistically appropriate communication and written/visual aids
	Family discussions are inconsistently documented	Standardised family-meeting record documenting

Section 6: Key Results – Objective 1

Shared Decision Making: Key findings (domains and themes) contributing to MCACP after evidence synthesis and evidence contextualisation after stakeholder interviews

Domain/theme	Evidence synthesis and stakeholder contextualisation	Contribution to MCACP
Decision-making capacity	Many critically ill patients cannot participate fully because of sedation, delirium, mechanical ventilation or neurological impairment. Capacity may fluctuate and is decision-specific.	Routine capacity assessment, repeated when necessary, with direct patient participation whenever possible
Surrogate identification	Families may be uncertain about who should represent the patient, and several relatives may express different views.	Early identification of the recognised surrogate/family spokesperson and documentation of their relationship and role
Patient values and preferences	Decisions are frequently centred on treatments rather than the patient's values, acceptable outcomes, prior wishes and lived experience.	Structured prompts to elicit what matters to the patient, unacceptable outcomes, treatment preferences and any advance directive
Information and deliberation	Families require understandable information about prognosis, uncertainty, benefits, burdens and reasonable alternatives before participating meaningfully.	A staged process of information sharing, values clarification, option comparison, deliberation and recommendation
Clinician–family roles	Families may either be excluded from decisions or feel burdened with sole responsibility for life-limiting choices. Stakeholders preferred shared responsibility.	Clinicians provide a values-informed recommendation while ensuring that families participate without carrying the entire decisional burden
Cultural and relational context	Indian ICU decisions are commonly family-centred and shaped by culture, spirituality, finances and collective responsibility. These factors must be acknowledged without overriding the patient's rights and interests.	Culturally responsive discussion, interpreter and psychosocial support, and explicit consideration of equity and financial vulnerability
Disagreement and review	Conflict may arise among clinicians, between relatives, or between the family and healthcare team, particularly when prognosis is uncertain.	Repeat meetings, second opinion, palliative-care consultation and escalation to a medical board/clinical ethics process when unresolved

Section 6: Key Results – Objective 1

Physician Consensus: Key findings (domains and themes) contributing to MCACP after evidence synthesis and evidence contextualisation after stakeholder interviews		
Domain/theme	Evidence synthesis and stakeholder contextualisation	Contribution to MCACP
Sources of disagreement	Differences in prognosis, perceived reversibility, specialty priorities and interpretation of appropriate treatment produce inconsistent recommendations and delays.	A structured multidisciplinary clinical review that separates areas of agreement, uncertainty and disagreement
Team alignment before family meetings	Families may receive conflicting messages when clinicians discuss treatment options without prior internal alignment.	Mandatory pre-family-meeting huddle and a single agreed clinical message, while transparently acknowledging uncertainty
Inclusive multidisciplinary participation	Decisions are often dominated by senior physicians. Nurses, palliative-care clinicians and allied-health professionals possess important information but may have limited influence.	Defined participation of the intensivist, primary treating team, nurse and palliative-care team, with allied-health input when relevant
Leadership and accountability	Unclear responsibility can result in repeated referrals, deferred decisions and diffusion of accountability between ICU and primary teams.	Identification of an accountable ICU lead, named communication lead, responsible documenter and specified action owners
Consensus under uncertainty	Complete unanimity may be impossible. Consensus should reflect a defensible shared recommendation rather than artificial certainty.	Record the majority/shared position, residual uncertainty, minority concerns, rationale and conditions for reassessment
Time-limited treatment trials	When benefit remains uncertain, an immediate all-or-none decision may be inappropriate. Stakeholders supported trial-based decisions with objective review.	A documented time-limited trial specifying treatment, duration, expected benefit, review criteria and action if goals are not achieved
Managing unresolved conflict	Persistent disagreement may occur within the clinical team or between clinicians and families, particularly around treatment limitation.	Stepwise escalation through repeat review, second opinion, palliative-care consultation, Primary/Secondary Medical Board and clinical ethics/legal support
Consensus under uncertainty	Verbal consensus may be lost across shifts or reinterpreted by	Standardised physician-consensus note recording participants,

Section 6: Key Results – Objective 1

Care Planning: Key findings (domains and themes) contributing to MCACP after evidence synthesis and evidence contextualisation after stakeholder interviews		
Domain/theme	Evidence synthesis and stakeholder contextualisation	Contribution to MCACP
Early anticipatory planning	Care planning often begins only after severe deterioration, leaving families and clinicians to make urgent decisions during crises.	Initiate Comprehensive Anticipatory Care Planning (CACP) when MCACP triggers are identified—not only during the final hours or days
Goals and values	Treatment plans may describe procedures without stating the patient’s goals, values, acceptable outcomes or preferred balance between longevity and comfort.	Document the overall goal of care, patient values, acceptable outcomes and treatment priorities
Treatment ceilings	Lack of explicit escalation plans contributes to repeated decisions, unwanted intervention and inconsistent care across shifts.	Specify decisions regarding CPR, intubation, ventilation, vasopressors, dialysis, surgery, investigations and other escalation
Time-limited treatment trials	When benefit remains uncertain, treatment may continue without predefined criteria for success, failure or review.	Record the intervention, duration, intended benefit, measurable review criteria and agreed action if treatment goals are not achieved
Symptom and supportive-care planning	Treatment-limitation documentation may focus on interventions being withheld while overlooking active comfort care and family support.	Include anticipatory plans for pain, dyspnoea, agitation, delirium, secretions, psychosocial and spiritual distress, and family support
Continuity across teams and settings	Plans may be lost during shift changes or transitions from ICU to ward, palliative-care unit, hospice or home.	A visible bedside action plan, structured handover and transition plan specifying responsible clinicians and receiving services
Review and revision	Goals and preferences may change with clinical improvement, deterioration or new information. Care plans should not be treated as permanent decisions.	Mandatory review after clinical change and at predefined intervals, with every revision dated, explained and communicated
	Incomplete documentation creates ethical and legal	Standardised CACP template documenting participants,

Section 6: Key Results – Objective 1

Care Delivery: Key findings contributing to MCACP after evidence synthesis and stakeholder contextualisation

Domain/theme	Evidence synthesis and stakeholder contextualisation	Contribution to MCACP
Early identification and activation	Palliative-care delivery is frequently delayed because needs are recognised late and referrals depend on individual clinicians.	Screen at ICU admission and reassess after deterioration, prolonged stay or escalating organ support; activate MCACP through needs-based triggers.
Integrated ICU–palliative-care model	Effective delivery combines primary palliative care by the ICU team with specialist support for complex needs.	Define responsibilities for ICU clinicians, nurses and specialist palliative-care teams, with clear referral and escalation pathways.
Symptom assessment and management	Pain, dyspnoea, agitation, delirium, secretions and psychological distress may remain under-recognised or inconsistently treated.	Use structured symptom assessment, anticipatory prescribing, treatment algorithms and documented reassessment of response.
Nursing-led continuity	Nurses provide continuous bedside assessment and recognise changes in symptoms and family distress, but their contribution is not always formalised.	Give nurses explicit responsibility for symptom screening, comfort measures, escalation, family support and care-plan continuity.
Bedside implementation of the care plan	Agreed treatment goals may not translate reliably into orders and actions across teams and shifts.	Convert the Comprehensive Appropriate Care Plan into visible clinical orders, a bedside action plan and structured handover prompts.
Care during the dying phase	Recognition of dying and transition from disease-directed treatment to active comfort care may be delayed or inconsistent.	Include dying-phase recognition, review of non-beneficial interventions, comfort-focused monitoring, medication guidance and family preparation.
Transitions and continuity of care	Patients may experience fragmented care during transfer from the ICU to a ward, palliative-care unit, home or another facility.	Use a standard transition checklist, portable care plan, medication and equipment plan, referral linkage and named follow-up responsibility.
Family and psychosocial support	Families require practical, emotional, cultural, spiritual and bereavement support alongside clinical information.	Embed social-work, psychological and spiritual-care assessment, family presence, culturally responsive support and bereavement follow-up.

Next Steps involved in Completing Objective 1

1. Freeze Initial MCACP v1.0

- Finalise the contents register
- Link programme theory, seven components, SOPs, algorithms, templates, training resources and workflows
- Assign version number and approval record

2. Complete NGT Documentation

- Consolidate participant details and meeting records
- Document priorities, disagreements and decisions
- Maintain a transparent consensus trail separate from the Delphi process

3. Conduct Two-Step Delphi

- Finalise multidisciplinary panel and consensus thresholds
- Round 1: rate components, core functions and tools
- Revise items with controlled feedback
- Round 2: confirm consensus, stability and retained items

4. Undertake Frontline Usability Testing

- Test the manual and tools with ICU physicians, nurses and palliative-care teams at both pilot sites
- Record task completion, completion time, errors and item-level feedback
- Measure acceptability, appropriateness and feasibility using AIM, IAM and FIM - ongoing

5. Finalise TIDieR and Adaptation Records

- Document materials, procedures, providers, setting, mode, dose and tailoring
- Classify each modification as affecting core function or operational form
- Update the version-controlled adaptation log

6. Release MCACP v1.1 and Close Objective 1

- Integrate Delphi, NGT and usability findings
- Finalise the MCACP manual and implementation toolkit
- Obtain Steering Committee approval
- Archive the complete evidence package and formally report Objective 1 closure

Section 6: Key Results – Objective 2

KMC Manipal: interim analysis cohort, n = 79	
Study profile	Findings
Recruitment and analysis cohort	79 of 100 planned participants recruited and included in the interim analysis by 31 August 2026
Participant characteristics	Mean age: 59.0 ± 14.5 years; male: 54.4%; rural residence: 49.4%
Illness trajectory	End-stage organ failure was the most frequent trajectory: 49.4%
Organ support	Mechanical ventilation: 86.8% among participants with available baseline support data; vasopressors: 34.2%
Palliative-care integration	Specialist palliative-care take over/admission under PC documented in 7.6%
Communication and care planning	Prognosis discussion documented in 10.1%; family meeting in 8.9%; advance directive in 2.5%
Family satisfaction	Median FS-ICU: 66.3 (IQR 53.8–76.0)
Quality of dying and death	QODD completed for 29 participants; median 22.6 (IQR 15.6–27.1)
Outcomes, costs and utilisation	ICU mortality: 36.7%; DAMA: 17.7%; median hospital bill: ₹4.43 lakh

Section 6: Key Results – Objective 2

SJMCH Bengaluru: 43 recruited; interim analysis cohort	
Study profile	Findings
Recruitment and analysis cohort	43 of 100 planned participants recruited by 31 August 2026
Participant characteristics	Demographic characteristics have not yet been included in the consolidated interim analysis
Illness trajectory	Illness-trajectory distribution has not yet been included in the consolidated interim analysis
Organ support	Organ-support data have not yet been included in the consolidated interim analysis
Palliative-care integration	Palliative-care referral documented in 70.0%; specialist consultation in 63.3%
Communication and care planning	Prognosis discussion: 100%; family meeting: 70.0%; goals-of-care documentation: 53.3%; treatment limitation: 60.0%; advance directive: 0%
Family satisfaction	Median FS-ICU: 65.2 (IQR 52.0–76.0)
Quality of dying and death	QODD completed for 10 participants; median 41.2 (IQR 33.1–62.1)
Outcomes, costs and utilisation	Median ICU cost: ₹2.59 lakh

Section 6: Key Results – Objective 2

Qualitative Study – Site 1		
Key theme	Site-specific finding	Implication for MCACP
Fragmented responsibility	The open ICU model involves multiple treating teams, resulting in variable ownership and inconsistent plans	Identify an accountable clinical lead and conduct multidisciplinary pre-family huddles
Prognostic uncertainty	Uncertainty can delay recognition of palliative-care needs and goal-setting	Use early screening triggers, serial prognostic review and time-limited treatment trials
Communication variability	Family discussions may occur late, be inconsistent or depend on individual clinicians	Nominate a communication lead; use structured, repeated family meetings and teach-back
Nursing participation	Nurses possess important bedside knowledge but are often peripheral to formal decisions	Define an explicit nursing role in assessment, team meetings and care planning
Documentation gaps	Prognosis, family discussions and treatment-limitation decisions are inconsistently recorded	Introduce standard documentation prompts and a visible bedside care plan
Palliative-care integration	Referral is frequently reactive rather than based on identified need	Introduce needs-based referral triggers and closer ICU–palliative-care collaboration
Ethical and legal concerns	Uncertainty regarding limitation of life-sustaining treatment can delay decisions	Provide institutional policy support, escalation pathways and clear documentation

Section 6: Key Results – Objective 2

Qualitative Study – Site 2		
Key theme	Site-specific finding	Implication for MCACP
Centralised clinical leadership	The closed, intensivist-led ICU model supports clearer responsibility and more consistent decision-making	Integrate MCACP review into consultant-led rounds and multidisciplinary meetings
Existing palliative-care access	Specialist palliative-care support is available, but referrals remain dependent on clinician recognition and discretion	Introduce needs-based referral triggers and defined ICU–palliative-care pathways
Early identification	Advanced or terminal illness may still be recognised late during the ICU admission	Incorporate screening at admission and scheduled reassessment
Team communication	Internal team alignment facilitates consistent communication, although structured pre-family discussions are not routine	Use brief team huddles before major family meetings
Family engagement	Families value honest, repeated and culturally sensitive discussions, particularly when prognosis is uncertain	Use structured family meetings, teach-back and explicit values exploration
Nursing contribution	Nurses provide continuity and detailed knowledge of symptoms and family concerns	Formalise nursing participation in assessment, meetings and care planning
Documentation	Prognosis, treatment goals and agreed limitations of treatment are not always documented consistently	Use standardised care-planning and family-meeting documentation templates

Section 6: Key Results – Triangulation of Results

Prospective Observational Study (Sites 1 and 2) + Qualitative Study (Sites 1 and 2)

How do they complement or contrast. How sites 1 and 2 are similar or differ

Domain	Site 1: KMC Manipal	Site 2: SJMCH Bengaluru	Triangulated interpretation
ICU organisation	Open ICU; care distributed across multiple specialty teams	Closed, intensivist-led ICU; clearer central responsibility	Governance structure influences ownership, consistency and speed of decision-making
Palliative-care integration	Limited routine integration; consultation largely reactive	Existing palliative-care access and closer ICU collaboration	Access alone is insufficient—both sites require systematic, needs-based referral triggers
Identification of need	Low documentation of palliative consultation, prognosis and family meetings	Recognition is more facilitated but remains clinician-dependent	Both datasets support early screening and scheduled reassessment
Communication	Greater risk of inconsistent messages from different treating teams	Better team alignment, but family meetings remain variably structured	A communication lead, pre-family huddle and repeated structured discussions are required
Care planning	Fragmented ownership may delay consensus and documentation	Centralised leadership facilitates planning, although documentation gaps remain	Standard care-planning and treatment-limitation documentation should be used at both sites
Nursing role	Bedside knowledge is not consistently included in formal decisions	Greater continuity, but nursing participation still requires formalisation	MCACP should explicitly include nurses in assessment, meetings and follow-up
Implementation requirement	Flexible workflow, accountability and cross-team coordination	Integration with established ICU–palliative-care processes	MCACP needs a common core with site-specific adaptations

Section 6: Key Results – Contribution of Objective 2 Results to Objective 1

MCACP intervention	Contribution of Objective 2 results to iterative development
Prognostication	Confirmed the need for admission screening, explicit eligibility triggers, serial prognostic reassessment and time-limited treatment trials when prognosis remains uncertain.
Communication	Informed a structured, culturally and linguistically responsive communication process involving a designated communication lead, repeated family meetings, scenario-based explanations and checks of understanding.
Shared Decision-Making	Demonstrated the importance of eliciting patient values, assessing decision-making capacity, identifying an appropriate surrogate and linking treatment options explicitly to the patient's goals and preferences.
Physician Consensus	Led to the inclusion of a multidisciplinary pre-family huddle to establish shared prognostic understanding, clarify clinical responsibility and agree on a consistent message and treatment recommendation.
Care Planning	Supported a standardised Comprehensive Appropriate Care Plan documenting goals, treatment ceilings, escalation or de-escalation decisions, symptom-management priorities and review timelines.
Care Delivery	Emphasised early needs-based palliative-care involvement, active symptom management, nursing participation, structured handovers and routine audit-and-feedback to improve

Section 7: Milestones Planned Vs Achieved

Obj .	WP / milestone	Planned milestone / deliverable	Due date	Achieved by Month 9	Status	Variance / explanation
1	WP1	Complete six realist reviews and develop initial program theory	14 Mar 2026	Six reviews completed; 466 studies synthesised; initial CMO configurations and programme theory developed	Completed	Achieved
1	WP2	Prepare initial MCACP manual/toolkit: SOPs, algorithms, templates, workflows and educational resources	14 Mar 2026	Seven-component working package developed; SOPs, CRFs, consent documents, templates, workflows and educational materials prepared; consolidated manual remains under iterative refinement	Substantially achieved (80–90%)	Minor carry-forward: final field-tested manual belongs to Objective 2/WP8
1	WP3	Undertake multidisciplinary stakeholder engagement and participatory co-design	14 Mar 2026	Consultations/FGDs undertaken with physicians, nurses, administrators/public-health experts, allied professionals and caregivers; outputs incorporated into MCACP	Substantially achieved (~85%)	Iterative user feedback continues during pilot by design

Section 7: Milestones Planned Vs Achieved

Obj.	WP / milestone	Planned milestone / deliverable	Due date	Achieved by Month 9	Status	Variance / explanation
2	WP4	Complete ethics approvals, site initiation, implementation teams and workflow preparation at two pilot ICUs	Pilot window ends 14 Mar 2027	Ethics/site initiation completed at KMC and SJMC; teams established; orientation and workflow mapping completed	Completed (100%)	Achieved
2	WP5	Complete baseline mixed-methods and organizational assessment	14 Mar 2027	142 patients screened; 122 recruited (KMC 79, SJMC 43); 66 qualitative interviews completed (KMC 35, SJMC 31); organizational assessment continues	In progress (~50–60%)	On schedule; objective only 3 months into 12-month pilot window
2	WP6	Pilot MCACP at KMC and SJMC and optimise workflow integration	14 Mar 2027	Pilot initiated at both sites; workflow testing, implementation workshops and iterative learning underway	In progress (~45–50%)	On schedule
2	WP7	Evaluate feasibility, acceptability, adoption, fidelity, barriers/facilitators and workflow integration	14 Mar 2027	Qualitative work and early process evaluation identify key barriers/facilitators; observations, fidelity assessment and implementation evaluation ongoing	In progress (~50–60%)	On schedule; final estimates require adequate implementation exposure
2	WP8	Refine MCACP manual/toolkit using pilot findings	14 Mar 2027	Priority refinements identified for prognostication, physician consensus, communication, family meetings, documentation, referral triggers and care evaluation	In progress (~40–50%)	On schedule; refinement correctly reported under Objective 2
	WP9	Develop dissemination and implementation plan	14 Mar 2027	Multicentre network established; early readiness	Initiated	On schedule; formal

Section 7: Milestones Planned Vs Achieved

Obj.	WP / milestone	Planned milestone / deliverable	Due date	Achieved by Month 9	Status	Variance / explanation
3	Multicentre implementation	Implement in four ICUs with palliative care access and evaluate effectiveness, implementation, cost and resource use	14 Sep 2028	Preparatory network and resource development initiated; formal implementation not started	Not yet due	Starts 15 Sep 2026 (Month 12)
4	Multicentre implementation	Implement in four ICUs without palliative care access; document effectiveness and low-resource adaptations	14 Sep 2029	No formal activity required during reporting period	Not yet due	Starts 15 Sep 2027 (Month 24)
5	Technical Resource	Deliver operational national hub containing peer-reviewed manuals, SOPs, videos,	14 Sep	Website active; educational, legal and implementation content under	Preparatory work	Formal objective starts 15 Sep 2027

Section 8: Objective-wise Deliverable Status

Objectives	Deliverables
Objective 1	Substantially completed: Six realist evidence reviews undertaken across the MCACP domains; stakeholder interviews completed; evidence synthesis and contextualisation underway. Draft MCACP components, programme theories and implementation tools developed for stakeholder validation and finalisation.
Objective 2	In progress: Baseline prospective observational and qualitative studies initiated at KMC Manipal and SJMCH Bengaluru. Recruitment, transcription and analysis are progressing; findings are being triangulated to refine the MCACP and prepare for feasibility testing at both sites.
Objective 3	Not yet due: Expansion to four ICUs with access to palliative care but without routine ICU-based delivery. Deliverables will include contextual assessment, implementation planning, implementation evaluation and effectiveness assessment.
Objective 4	Not yet due: Extension to four ICUs without established palliative-care access. Deliverables will include readiness assessment, local capacity-building, adapted implementation strategies and evaluation of implementation and clinical outcomes.
Objective 5	Preparatory work initiated: Technical Resource Hub concept and proposed resource domains identified. Final deliverables will include MCACP manuals, clinical tools, training materials, implementation guidance, audit indicators and an online knowledge-support platform.

Section 8: Spin-off Deliverables - academics

Spin-off Project	Contribution to the CAR
Implementing a critical care review board in Indian ICUs: an EPIS-guided implementation research study (PhD)	The study will develop evidence-based governance structures, operational policies, accountability frameworks, and decision-making processes that will strengthen the physician consensus component of the MCACP and support wider implementation across Indian ICUs.
ICU Nursing Education in Palliative Care (PhD)	Findings will inform the development of competency-based educational modules, training resources, and implementation strategies for ICU nurses, strengthening workforce capacity within the MCACP.
ICU Communication as a Dynamic Field: A Gestalt Analysis of Clinician-Family Interactions (PhD)	Findings will refine communication frameworks, family meeting guides, and shared decision-making strategies within the MCACP by providing a deeper understanding of the interactional processes that occur during serious-illness conversations.
Assessment of ICU Readiness for Implementing a Multifaceted Palliative Care Intervention: A Mixed-Methods Study (MD Thesis)	The study will identify organisational determinants of successful implementation and contribute to the development of readiness assessment tools, implementation checklists, and site-preparation strategies for the future multicentre rollout of the MCACP.
Evaluating the Acceptability, Feasibility, and Fidelity of Early Implementation of the MCACP Communication and Prognostication Bundle in the ICU: A Prospective Observational Study (MD Thesis)	The study will generate implementation outcome data on acceptability, feasibility, fidelity, and workflow integration, allowing refinement of intervention delivery and informing the implementation strategy for the multicentre phase of the CAR programme.
Developing, Implementing and Evaluating an Integrated, Co-Designed Health Resource Utilisation and Economic Evaluation Framework for People with Advanced Life-Limiting Illnesses in Indian ICUs (PhD)	This PhD research focuses on creating and testing a healthcare framework that measures how medical resources are used and what they cost when caring for critically ill patients with advanced life-limiting conditions in Indian Intensive Care Units (ICUs). It aims to support more efficient, patient-centered, and economically sustainable healthcare decision-making.
Conversations on Future Care Preferences during Family Meetings in a Critical Illness Setting: A Discursive Psychological Analysis (MD Thesis)	This MD thesis examines how patients, families, and healthcare professionals discuss future treatment and care preferences during family meetings in critical care settings. Using discursive psychology, it analyzes communication patterns to better understand how important care decisions are negotiated and expressed

Section 9: Outcomes - Publications

Manuscript	Title
Published Manuscripts	- Vallath N, Parvathy V, Padhi P, Johny T A, Krishnakumar M, Varghese PS. Procedures to implement the Supreme Court of India directives to withdraw/withhold life-sustaining medical treatment. Indian J Med Res. 2026 Mar;163(3):319-326. doi: 10.25259/IJMR_3005_2025 - Ghoshal, A., Mathew, M., Jain, V. et al. Integrating Specialist Palliative Care into ICU workflows in India: a six-year theory-informed evaluation of the BLUE MAPLE program. BMC Palliat Care (2026). https://doi.org/10.1186/s12904-026-02295-7 - Ghoshal A, Prasad A, Nanda A, Salins N. How prognostic information influences care planning in adult intensive care units: protocol for a realist review. BMJ Open. 2026;16(6):e117449. Published 2026 Jun 29. doi:10.1136/bmjopen-2026-117449
Manuscripts submitted and under peer-review	Systematic Reviews journal - Protocol for a realist review of palliative care delivery in intensive care units: mechanisms influencing symptom assessment and management and grief/bereavement support - Protocol: Realist Review of Shared Decision-Making in Intensive Care Units - Communication during high-stakes ICU family-clinician interactions: a realist review protocol BMJ Open journal - Structured ICU Governance Mechanisms to Strengthen End-of-Life Decision-Making: A Realist Review Protocol - Advance Care Planning in Intensive Care Units for People with Serious Life-Limiting Illnesses: A Realist Scoping Review Protocol
Manuscripts under preparation	Realist reviews – Prognostication, Communication, Shared Decision-Making, Physician Consensus / Multidisciplinary Approach, Care Planning, Care Delivery (n=6)

Section 9: Outcomes – training/ workshops

Training program/ workshop	Title
Training of nurses and health care workers working in ICUs - KMC	Every Saturday between 1400 to 1500 hrs
Qualitative research methods workshop – SJMC (2026)	Expert facilitators from NIMHANS, SJMCH, SJRI, University of Pittsburgh, USA Key training areas: - Qualitative research paradigms & approaches in critical care - Developing qualitative research protocols - Data collection: Interviews & Focus Group Discussions (FGDs) - Rigor, ethics & quality in qualitative inquiry - Thematic analysis: coding, categories & theme development - Digital tools: Introduction to qualitative research software & hands-on NVivo coding Training methods: Interactive lectures, discussions, practical exercises & hands-on sessions
Implementation research Workshop – SJMC (2026)	Co-hosted by: Departments of Palliative Medicine & Critical Care Medicine, SJMCH Expert faculty: ICMR, KMC Manipal, SJMCH, SJRI, University of Pittsburgh & national/international experts Participants: Students, researchers, nurses, clinicians and faculty Core focus: Implementation Science concepts, frameworks and evidence-to-practice translation CFIR: Identifying barriers and facilitators across multiple levels ERIC & RE-AIM: Linking implementation barriers with appropriate implementation strategies Practical learning: Case studies, group activities, Think-Pair-Share and team-based exercises Implementation planning: Developing context-specific strategies, measures and evaluation plans Application: Strengthening readiness to integrate palliative care into ICU practice

Section 9: Outcomes – documents/ administrative changes developed for introducing PC upstream

Document	Purpose
Non-escalation-related	SJMC FORM 1 screening and enrolment baseline and admission characteristics initial care needs and clinical status the pathway to non-escalation, patient/family confidence and comfort with the decision. FORM 2 the quality of inpatient palliative care the processes and support involved in transitioning patients to their preferred place of care – if any continuity of care and patient outcomes during follow-up beyond discharge. FORM 3 family satisfaction, understanding, preparedness, decision-making experience, emotional responses Experience of opting for non-escalation, consequences. FORM 4 Direct and indirect costs healthcare resource utilization financial impact of non-escalation and palliative care on patients and families. KMC – the BLUE MAPLE document
Goals of care document/ processes	SJMC Constitution of the primary, secondary medical boards (PMB, SMB) and reporting formats for PMB, SMB evaluations has been accomplished KMC – Critical care review board
living will worksheet template	SJMC 1.SOPs for Transitioning ICU patient with guarded prognosis to Palliative Care ApTCare-ICU–Criteria for Appropriate Transitions in Care for person with serious illness - Applied at / after 2 days of ICU 2.SOPs for compassionate extubating and EOL C within the ICU

Section 10: Proposed Workplan 2026-27

Objectives	Workplan 2026-27
Objective 1	Complete transcription and thematic synthesis; triangulate realist-review and stakeholder findings; finalise program theories and MCACP components; conduct expert/stakeholder validation; freeze MCACP version 1.0, implementation manual and associated clinical tools.
Objective 2	Complete baseline observational studies at both pilot sites; undertake site-specific contextual assessment; train ICU and palliative-care teams; introduce MCACP through iterative cycles; evaluate adoption, feasibility, acceptability, fidelity and preliminary clinical outcomes; refine MCACP to version 2.0.
Objective 3	Identify and onboard four ICUs with access to palliative care but without routine ICU-based delivery; complete regulatory approvals and baseline ecosystem assessments; map barriers and facilitators; co-design site-specific implementation strategies and initiate phased implementation.
Objective 4	Finalise selection and readiness assessment of four ICUs without palliative-care access; establish mentoring and referral linkages; assess workforce and training requirements; develop contextual adaptations and capacity-building plans for subsequent implementation.
Objective 5	Develop the Technical Resource Hub architecture; curate evidence summaries, MCACP tools, training modules and implementation resources; develop fidelity and audit dashboards; conduct user testing and launch an initial prototype for participating centres.

Thank
you

By Team MATHRU